Ware2Door

Database Design Documentation

ER Diagram · Normalisation · Relation Schema

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1. Entity-Relationship Diagram

The ER diagram below represents the conceptual data model for the Ware2Door logistics system. It captures all entities, relationships, cardinalities, and participation constraints.

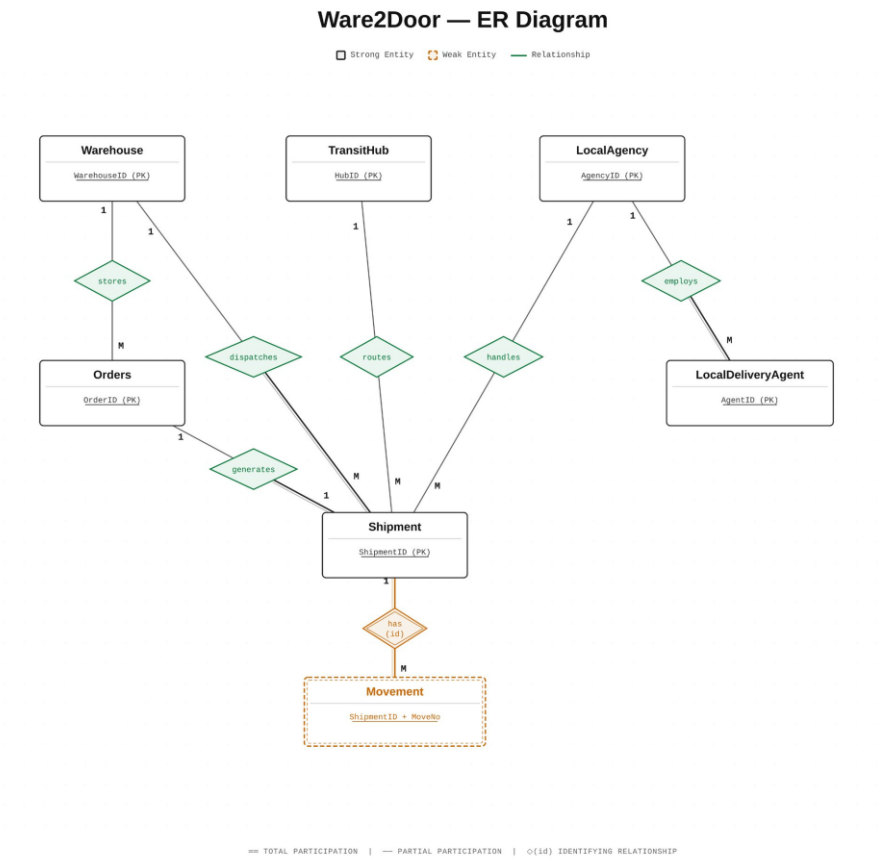

Figure 1: Ware2Door ER Diagram

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2. Normalisation

Normalisation is the process of organising a relational database to reduce data redundancy and improve data integrity. The following section documents the step-by-step normalisation applied to the Ware2Door database schema.

2.1 First Normal Form (1NF)

Definition

A relation (table) is in First Normal Form (1NF) if:

- All attributes contain only atomic (indivisible) values.
- There are no repeating groups or multi-valued attributes.

Application to Ware2Door

We identified attributes that could be divided into further atomic states and decomposed them accordingly:

Example 1 — Location Attribute

The Location attribute in WareHouse, TransitHub, and LocalAgency is a composite attribute that can be broken down into atomic components:

- Location → address, city, state, pincode

Example 2 — Geometric Coordinates in Movement

The Geometric_Coordinates attribute in Movement stores both latitude and longitude as a single value. This violates 1NF and was divided into separate atomic attributes:

- Geometric_Coordinates → latitude, longitude

2.2 Second Normal Form (2NF)

Definition

A relation (table) is in Second Normal Form (2NF) if:

- It is already in First Normal Form (1NF).
- Every non-prime attribute is fully functionally dependent on the entire primary key (no partial dependencies).

Application to Ware2Door

After completing 1NF, we identified a partial dependency in our model:

Example — Local_Delivery_Agent Entity

In the original design, the Local_Delivery_Agent entity had a composite primary key of (agent_id, agency_id). However, the non-prime attributes agent_name, agent_phone, and agent_email were fully dependent only on agent_id, not on the full composite key — a clear partial dependency.

Resolution: The entity was decomposed so that agent_id became the sole primary key of Local_Delivery_Agent, and agency_id was retained as a foreign key to maintain the relationship with LocalAgency. This eliminates the partial dependency and brings the schema into 2NF.

2.3 Third Normal Form (3NF)

Definition

A relation (table) is in Third Normal Form (3NF) if:

- It is in Second Normal Form (2NF).
- There is no transitive dependency for non-prime attributes (i.e., no non-prime attribute depends on another non-prime attribute).

Application to Ware2Door

After converting to 1NF and 2NF, the Ware2Door schema was examined for transitive dependencies. No transitive dependencies were found — all non-prime attributes in every table depend directly and only on the primary key of their respective table.

Result: The schema satisfies 3NF without requiring any further decomposition.

2.4 Boyce–Codd Normal Form (BCNF)

Definition

A relation is in Boyce–Codd Normal Form (BCNF) if:

- For every non-trivial functional dependency X → Y, X must be a superkey.

Application to Ware2Door

After applying 1NF and 2NF, the Ware2Door database schema was directly promoted to full BCNF compliance. No additional transformations for 3NF or BCNF were necessary, as all functional dependencies in every table are determined by superkeys only.

Final Result: The Ware2Door schema is in BCNF.

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3. Relation Schema

The relation schema below shows the final normalised tables for the Ware2Door system, including all primary keys (underlined), foreign keys, and their relationships.

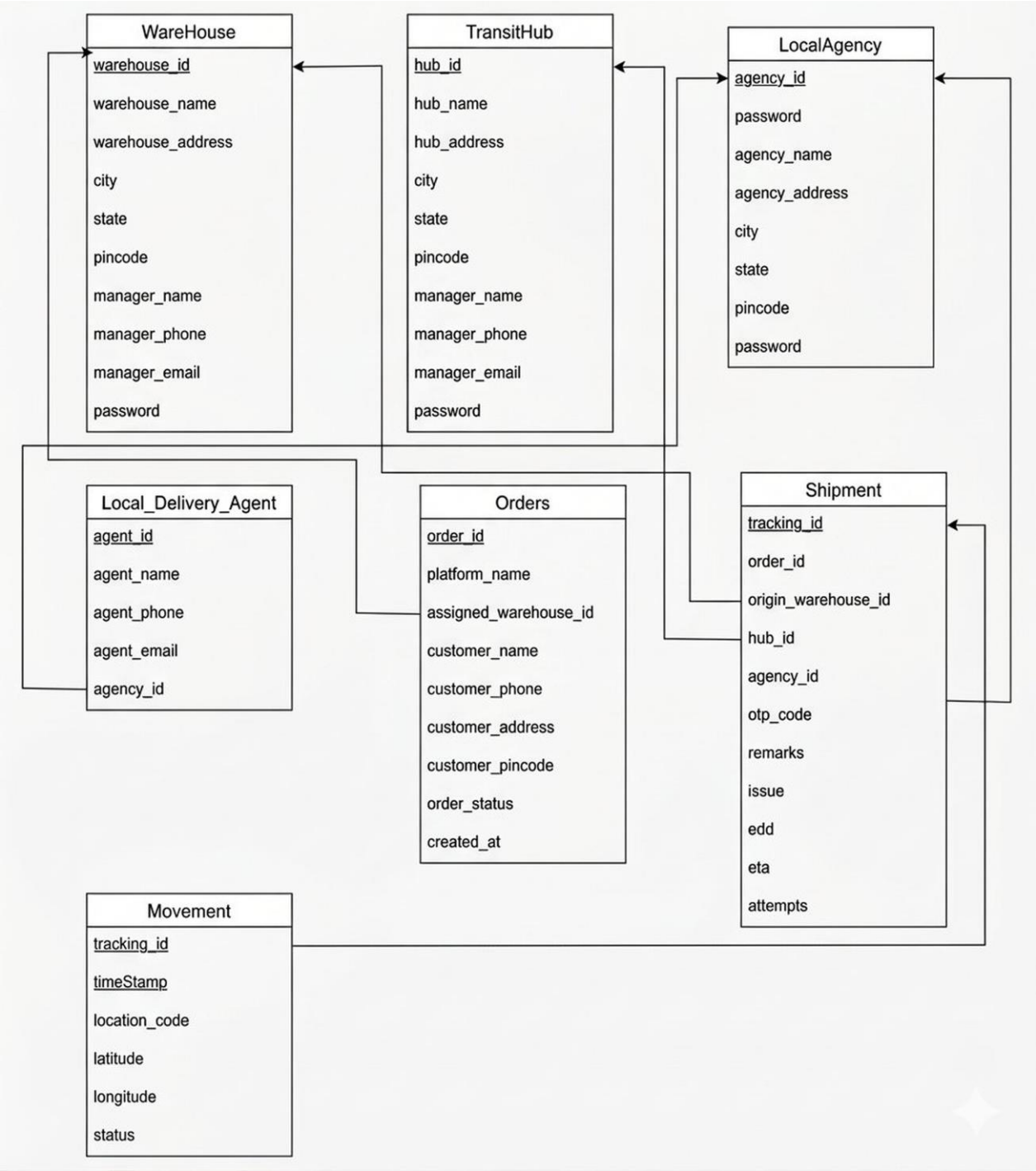

Figure 2: Ware2Door Relation Schema